Physical Distributed GNSS Spoofing of UAV Swarms: Formation Dilation, Collapse-Detector Blindness, and the Two-Sided Test That Detects It

Silvia Schilleci
[TODO: affiliazione, email]

***Abstract*—Multi-receiver spoofing detection relies on a geometric fact: a single-transmitter spoofer forces every receiver of a formation onto the same position, so the distances implied by the GNSS fixes collapse. We simulate, transmitter by transmitter, a *physical distributed* spoofer—24 fixed ground transmitters around the operating area of a 25-UAV swarm, one impersonated satellite each, with power-based capture through an elevation-dependent ground-to-UAV channel—over the 5–20 m spacing of GPS-based formation flight, 20 Monte Carlo missions per configuration. A ground array cannot preserve the formation: unable to reproduce the satellites' elevation, it *dilates* the computed formation by about 23 % of the baseline, as predicted by a validated first-order model. The dilation is invisible to the collapse detectors of Jansen et al. (ACSAC 2016), which test the wrong tail ($P_{\mathrm{D}} \leq 0.011$ at $P_{\mathrm{FA}} = 10^{-2}$), and is largely masked by noise for the worst-pair GNSS-vs-UWB check ($P_{\mathrm{D}} \leq 0.26$); a two-sided test on the median distance ratio detects it reliably from 15 m and in a majority of epochs from 8 m. Raising the transmitters turns the dilation into a contraction: the aggregate test is blind for geometric dilations within $[-8, +5]$ %, reached at a mission-dependent altitude around the zero crossing (7 km for a 10 km array), and a WASP-like constellation on the satellites' lines of sight at several kilometres of slant range is detected in fewer than a quarter of the epochs. The dilation signature, not the collapse, is what relative-distance detectors should test for against ground spoofers, and the altitude needed to erase it is the attacker's cost.**

***Index Terms*—GNSS spoofing, UAV swarms, spoofing detection, multi-receiver, distributed attack, relative distances, formation flight.**

I. Introduction

Civilian GNSS signals carry no authentication: their structure and navigation message are public, and a transmitter that reproduces them with enough power can make a receiver compute any position and time of the attacker's choosing. Since the first portable civilian spoofer was demonstrated [1], software-defined radios and open-source signal generators have made the attack accessible for a few hundred dollars, and incidents affecting ships, aircraft and unmanned systems are reported routinely [2]. Unmanned aerial vehicles (UAVs) are a natural target: they navigate on GNSS by default, and a swarm flying in formation—for surveying, inspection, delivery, or defense—multiplies both the value of the target and the consequences of a successful hijack.

Among the countermeasures proposed in the last fifteen years [2], [3], *multi-receiver* detection stands out for its simplicity and for the strength of its guarantee. Its principle, established by Tippenhauer et al. [4], is geometric: a spoofer with a single antenna delivers the same signals to every receiver in its coverage area, so all the victims compute the *same* position fix. A group of receivers that knows its own formation can therefore detect the attack by checking that the mutual distances implied by the GNSS fixes have not collapsed. Jansen et al. [5] turned the check into a calibrated statistical test, showing that four receivers a few metres apart reach a detection probability of 0.99 at a false-alarm rate of 10^{-5}; Swaszek and Hartnett [6] derived its Neyman–Pearson optimal form. For UAV swarms the same idea has been proposed with the formation measured on-line by inter-drone ultra-wideband (UWB) ranging [7] or verified as a localization feasibility problem [8]. The guarantee rests on one assumption: that the attacker has a single transmission point.

This paper asks what happens when the assumption is dropped in the most concrete way possible—not with an idealized attacker that imposes a common offset on every victim, but with a *physical distributed* spoofer: several fixed ground transmitters surrounding the swarm's operating area, each impersonating one satellite, subject to real propagation geometry and a realistic ground-to-UAV channel. Such an attacker is within reach of a well-resourced adversary, and formation-preserving spoofing has been demonstrated in hardware when the transmitters are placed along the satellite directions at altitude [9]. Ground-based transmitters, however, cannot reproduce the satellites' elevation; what they do to the computed formation, and which relative-distance test sees it, has not been quantified.

We answer by simulation over the inter-UAV spacing of 5 to 20 m that characterizes outdoor GPS-based formation flight (Section VI-A). Our contributions are:

- a transmitter-by-transmitter simulation model of a distributed ground spoofer attacking a moving UAV swarm, with per-epoch routing of satellites to transmitters, true three-dimensional propagation of the replayed signals, and the elevation-dependent ground-to-UAV path-loss model of [10] (Section IV);

- the finding, with a validated first-order model, that a ground array *dilates* the computed formation by about $1/\langle \cos \mathrm{el} \rangle - 1 \approx 23\,\%$ of the baseline, because of the mismatch between satellite and transmitter elevations (Section IV-C);
- a Monte Carlo evaluation with mission-level bootstrap confidence intervals of five relative-distance detectors—the three collapse variants of [5], the worst-pair GNSS-vs-ranging check [7], [11], and a two-sided test on the median distance ratio—showing that the dilation escapes the first four and is detected by the fifth at every spacing (Section VI);
- a transmitter-altitude sweep and a line-of-sight (WASP-like) placement experiment that measure the cost, in altitude, of erasing the dilation: the signed dilation crosses zero at about 36° of transmitter elevation (7 km for a 10 km array), the aggregate test is blind for geometric dilations within $[-8, +5]\,\%$, and a static constellation on the satellites' lines of sight at ≥ 5 km slant range is detected in fewer than a quarter of the epochs (Section VI-E);
- a delimitation of the result with respect to direction-of-arrival and absolute-reference defenses (Section VII).

The simulator and the detection code are released with the paper.

II. Related Work

A. Multi-receiver detection in the position domain

Tippenhauer et al. [4] formalized the GPS group-spoofing problem and proved two complementary results: a single-antenna spoofer places all victims at the same location (Result 1), and a formation-preserving placement of one transmitter per impersonated satellite always exists, along the satellite directions (Result 7). Their countermeasure checks that the computed positions preserve the physical formation, without a statistical model. Jansen et al. [5] supplied the model: the pairwise distances between receivers are Rician under authentic conditions and Rayleigh under a single-transmitter attack, with spatially correlated position noise ($\sigma \approx 0.5$–3 m, $\rho \approx 0.4$–0.6 measured), and the test $f(\{d_{ij}\}) < \lambda$ is evaluated in minimal, maximal, majority and normalized-majority variants, the last being the best. Their prototype with four receivers on a 5 m square ran for three hours without errors; their own attempt at a distance-preserving four-USRP attack succeeded on one receiver and failed on two. Swaszek and Hartnett [6] derived Neyman–Pearson optimal tests for COTS receivers at known relative positions, and later fused GNSS fixes with inter-vehicle ranges for platoons under the assumption that only one vehicle is spoofed [12]. Crowd-GPS-Sec [13] applies the collapse principle to crowdsourced air-traffic receivers, combined with an absolute time-difference-of-arrival reference.

B. Measurement-domain and swarm variants

In the measurement domain, Stenberg et al. [11] test double differences of pseudorange and carrier phase between receivers with a χ^2 GLRT, finding a minimum receiver separation of about 5 m with standard positioning; Kalantari and Larsson [14] derive a GLRT for receivers on a rigid body with unknown pose explicitly against multiple spoofers, showing that more receivers make the attack harder; Borio and Gioia [15] detect the single common direction of arrival with a two-receiver carrier-phase sum-of-squares statistic. For UAV swarms, Mykytyn et al. [7] compare GNSS-derived inter-drone distances with UWB two-way ranging and raise an alarm when they differ by more than about 5 m, dismissing a coordinated per-drone attack as "virtually impossible"; Bi et al. [8] cast detection as a semidefinite localization-feasibility problem robust to colluding members. GNSS/INS fusion defenses such as SemperFi [16] and tightly coupled INS innovation monitors [17], [18] compare the GNSS track with inertial dead reckoning and respond to the time variation of the induced error.

C. Coordinated and distributed spoofing

Attack-side work has progressively approached the formation-preserving placement of [4]. Ceccato et al. [19] designed a multi-antenna ground spoofer with Wiener precoding that displaces a swarm without disrupting its formation, at the level of a mathematical model; Chen et al. [20] analyze formation-preserving spoofing of small swarms with software-defined radios; GNSS-WASP [9] demonstrated in hardware a constellation of at least four GPSDO-synchronized transmitters, placed in line with the satellites at altitude, that spoofs an arbitrary number of receivers over a square kilometre while preserving their relative distances to within a metre, explicitly bypassing multi-receiver [5] and sensor-fusion [16] countermeasures; its authors note that the residual bias of a real transmitter constellation "cannot be eliminated unless they are placed in orbit". GNSS-WASP also discusses a WASP-aware countermeasure that accumulates the worst-pair residual over time (CUSUM), appropriate for a residual of variable sign; the detector we propose in Section V instead aggregates *spatially* over the pairs, which is possible because the residual of a ground array has a coherent sign (a dilation). Park and Yoo [21] state the translation-invariance of relative-only detectors as a formal proposition, rooted in the gauge freedom of anchor-free network localization [22], and evaluate it by simulation with an idealized common offset. Surveys of UAV spoofing defenses [3], [23] note that detectors relying on neighbours' positions fail when the neighbours are inside the spoofed area, but none quantifies the effect for a physical ground array.

D. Positioning of this work

Our attacker is deliberately more constrained than those of [9], [21]: fixed transmitters on the ground, no per-victim adaptation, a measured channel. The quantity we report is precisely what separates it from the ideal formation-preserving attack—a dilation of the computed formation, modeled and

validated—and its consequence for the relative-distance detectors over the spacing range of real formation flight, including the two-sided aggregate test that the dilation calls for.

III. System Model

A. Swarm, constellation, and measurement model

We consider a swarm of $m = 25$ UAVs flying in a square 5×5 grid with spacing g (the *formation spacing*), following a way-point trajectory at altitudes between 100 and 700 m that spans about 7 km around its start point; positions are sampled every $T_s = 5$ s over a 20-minute mission. The GNSS constellation is modeled as $N_{\mathrm{sat}} = 24$ satellites on 6 circular orbital planes (inclination 55°, altitude 20 200 km) over a rotating spherical Earth. For UAV u at position $\mathbf{p}_u(t)$ and satellite s at position $\mathbf{s}_s(t)$ (visible when its elevation is positive), the authentic propagation delay is $\tau_{u,s}(t) = \|\mathbf{s}_s(t) - \mathbf{p}_u(t)\|/c$. Each receiver has a clock bias $\delta t_u \sim \mathcal{U}(-1, 1)$ s, constant over the mission, and each pseudorange carries additive white Gaussian noise of standard deviation $\sigma_{\mathrm{pr}} = 3$ m, representative of standard (non-differential) positioning.

Each UAV computes its position fix $\hat{\mathbf{p}}_u$ independently by iterated linearized least squares (Gauss–Newton) over the four unknowns $(x, y, z, \delta t)$: at iteration k, with predicted ranges $R_s = \|\mathbf{s}_s - \hat{\mathbf{x}}^{(k)}\|$ and pseudoranges $\rho_s = c\,(\tilde{\tau}_{u,s} - \widehat{\delta t}^{\,(k)})$,

$$\mathbf{G}\,\Delta = \boldsymbol{\rho} - \mathbf{R}, \qquad \mathbf{G}_{s,:} = \left[\tfrac{(\hat{\mathbf{x}} - \mathbf{s}_s)^\top}{R_s},\ c\right], \tag{1}$$

solved in the least-squares sense for five iterations, where $\tilde{\tau}_{u,s}$ is the *measured* delay (authentic or spoofed). Without attack and noise the solver returns the true grid positions to numerical precision (error $< 10^{-4}$ m), so estimated and true positions are expressed in the same frame.

In addition to GNSS, the UAVs measure their instantaneous mutual distances with an independent on-board ranging sensor (UWB two-way ranging, as in [7]):

$$d_{ij}^{\mathrm{meas}}(t) = \|\mathbf{p}_i(t) - \mathbf{p}_j(t)\| + \varepsilon_{ij}(t), \quad \varepsilon_{ij} \sim \mathcal{N}(0, \sigma_{\mathrm{rng}}^2), \tag{2}$$

with $\sigma_{\mathrm{rng}} = 0.1$ m. This is the most favorable setting for the defender: the true formation is known essentially without error at every epoch.

B. Ground-to-UAV channel

Spoofing signals travel over a ground-to-air link. We adopt the suburban cellular-to-UAV path-loss model of Al-Hourani and Gomez [10]: for a ground transmitter at horizontal distance d and elevation angle θ (in degrees) as seen from the UAV,

$$\mathrm{PL}(d, \theta) = 10\,\alpha \log_{10}(d) + A\,(\theta - \theta_0)\,e^{-\frac{\theta - \theta_0}{B}} + \eta_0 \quad [\mathrm{dB}], \tag{3}$$

with $\alpha = 3.04$, $A = -23.29$, $B = 4.14$, $\theta_0 = -3.61^\circ$, $\eta_0 = 20.70$ dB (shadowing disabled for repeatability). The excess-loss term is fitted in [10] for $\theta \in [-2^\circ, 10^\circ]$ and is saturated at 10° beyond; with the deployment of Section IV the largest angle met in any run is 6.0°, so the model is always used within its validity range. Authentic signals follow free-space loss from satellite transmit power $P_t^{\mathrm{sat}} = 25$ W with antenna gains $G_t^{\mathrm{sat}} = 13.5$ dB and $G_r = 5$ dB, giving received powers of about -150 dBW.

IV. Physical Distributed Spoofing Attack

The attacker's goal is to displace the whole swarm's perceived position onto a decoy point about 5.7 km from the flight path, while the swarm believes it is following its mission plan. The attack is simulated transmitter by transmitter; nothing is assumed about per-victim signal shaping. As in [9], the spoofer replays the authentic navigation message, so the receivers use the true satellite ephemerides and only the code phases are forged.

A. Transmitter geometry and per-epoch routing

$N_{\mathrm{sp}} = 24$ fixed ground transmitters are placed on a circle of radius $R_{\mathrm{sp}} = 10$ km enclosing the swarm's operating area (Fig. 2), each radiating $P_t^{\mathrm{sp}} = 1$ mW with $G_t^{\mathrm{sp}} = 1$ dB. Following the geometric argument of [4, Result 7], each visible satellite s is routed, at every epoch, to the transmitter whose azimuth as seen from the *current* true swarm centre $\mathbf{p}^{\mathrm{ctr}}(t)$ is closest to the satellite's azimuth:

$$\mathrm{sp}(s, t) = \arg\min_k \left|\angle_{2\pi}\big(\phi_s(t) - \phi_k(t)\big)\right|. \tag{4}$$

The transmitters never move; only the routing is updated, which requires the attacker to track the swarm centre—the same knowledge it needs to calibrate the replayed delays.

B. Capture and replayed delay

For each UAV u and visible satellite s the receiver locks onto the transmitter whenever the received spoofing power, computed with (3), exceeds the authentic one. With the chosen EIRP the spoofer margin over the satellite is between 13 and 29 dB on every link (median 19.5 dB), so capture is complete and homogeneous across the swarm: no UAV mixes authentic and spoofed satellites.

Transmitter $k = \mathrm{sp}(s, t)$ replays the signal of satellite s with its code phase calibrated so that, at the true swarm centre, the measured delay equals the decoy delay τ_s^{dec}—the delay that satellite s would exhibit at the decoy position. A UAV at a different position receives the same transmission over a different path and measures

$$\tilde{\tau}_{u,s} \;=\; \tau_s^{\mathrm{dec}} \;-\; \delta t_u \;+\; \frac{\|\mathbf{q}_k - \mathbf{p}_u\| - \|\mathbf{q}_k - \mathbf{p}^{\mathrm{ctr}}\|}{c}, \tag{5}$$

where $\mathbf{q}_k$ is the transmitter position and both norms are true three-dimensional slant distances. The last term is the physical propagation residual: it vanishes for the centre UAV, which computes exactly the decoy position, and is nonzero for the other UAVs. Since the delays in (5) do not depend on the transmitted power, the residual is by construction independent of it once capture is complete.

C. Residual non-rigidity: a ground array dilates the formation

Write $\mathbf{p}_u = \mathbf{p}^{\mathrm{ctr}} + \boldsymbol{\delta}_u$ with $\|\boldsymbol{\delta}_u\| \ll \|\mathbf{q}_k - \mathbf{p}^{\mathrm{ctr}}\|$, and let $\hat{\mathbf{v}}_k$ be the unit vector from the centre to transmitter k and $\hat{\mathbf{e}}_s$ the unit line of sight from the centre to satellite s. To first order the delivered residual in (5) is $-(\hat{\mathbf{v}}_k \cdot \boldsymbol{\delta}_u)/c$, whereas the delay that would make UAV u compute the decoy position *rigidly*

displaced by $\boldsymbol{\delta}_u$ is $-(\hat{\mathbf{e}}_s \cdot \boldsymbol{\delta}_u)/c$. The pseudorange error with respect to the rigid attack is therefore

$$\epsilon_{u,s} \;=\; (\hat{\mathbf{e}}_s - \hat{\mathbf{v}}_k) \cdot \boldsymbol{\delta}_u \;\approx\; -(\hat{\mathbf{a}}_s \cdot \boldsymbol{\delta}_u)\,\big(\cos\theta_k - \cos \mathrm{el}_s\big), \quad (6)$$

where the approximation holds for a horizontal swarm and azimuth-matched routing, with $\hat{\mathbf{a}}_s$ the satellite azimuth direction, θ_k the near-zero elevation of the ground transmitter and el_s the satellite elevation. Since $\cos\theta_k \approx 1 > \cos \mathrm{el}_s$, the coefficient is negative for every satellite: a UAV displaced *towards* the azimuth of satellite s measures a pseudorange *shorter* than the rigid one, and a UAV displaced away from it a longer one. The least-squares solver (1) interprets shorter ranges to the satellites ahead and longer ranges to those behind as a position farther along $\boldsymbol{\delta}_u$: every UAV is pushed outward along its own displacement from the centre, and the computed formation is *dilated*. A ground transmitter cannot reproduce the satellite's elevation, so (6) vanishes only for satellites near the horizon.

Predicted dilation. Matching (6) with the linearized position response $-\hat{\mathbf{e}}_s \cdot \Delta\mathbf{p}$ gives, for a single satellite, $\Delta\mathbf{p} = (1/\cos \mathrm{el}_s - 1)\,\boldsymbol{\delta}_u$; averaged over the constellation the dilation is of order

$$E \;\approx\; \frac{1}{\langle \cos \mathrm{el}_s \rangle} - 1, \quad (7)$$

which gives 0.3–0.4 for typical elevation distributions and is a useful upper-bound intuition; the exact first-order prediction is obtained by solving the same least-squares problem (1) with the error vector $\{\epsilon_{u,s}\}$ for a unit displacement, which yields a linear map $\mathbf{M}$ from $\boldsymbol{\delta}_u$ to the position error (including its vertical component) and hence the relative length change of every pair of the grid. Figure 1 compares this prediction with the measured distortion, epoch by epoch, on a noiseless attacked mission at $g = 20\,\mathrm{m}$. The agreement (median predicted $24.4\,\%$ against measured $24.5\,\%$, correlation 1.00, largest deviation 0.4 percentage points) is essentially a check that the first-order linearization holds at these baselines; its value lies in what the comparison of the three predictions shows: the simple formula (7) gives $34.5\,\%$, the horizontal part of the linear model $18.8\,\%$, the full model $24.4\,\%$. The clock and vertical degrees of freedom of the solver absorb part of the pseudorange error (hence $34.5 \to 24.4$), and the vertical component of the position error—an altitude tilt proportional to $\boldsymbol{\delta}_u$—adds a quarter of the three-dimensional distortion (hence $18.8 \to 24.4$). The variation of the dilation with the elevations of the satellites in view is also what produces the mission-to-mission spread of Fig. 4.

Two further facts matter for an attacker. The residual does not depend on transmitter power, by construction. It depends critically on the array enclosing the operating area: with a $2.5\,\mathrm{km}$ array centred on the start point the swarm is inside the circle only $26\,\%$ of the time, and the deformation grows from $12\,\%$ inside to $55\,\%$ outside (correlation 0.98 with the distance from the array centre), because from outside the array all transmitters lie in a narrow angular sector and no routing can match the satellite azimuths.

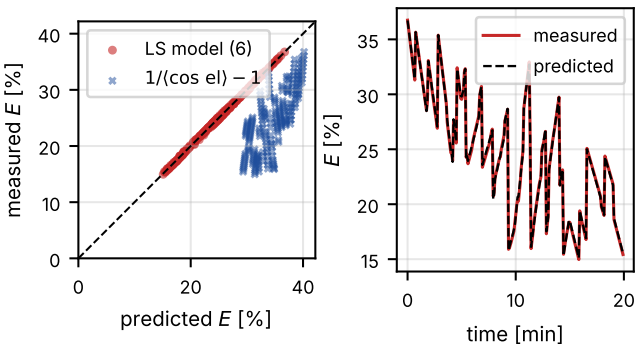


Fig. 1. Validation of the dilation model on a noiseless attacked mission ($g = 20\,\mathrm{m}$, 240 epochs). Left: measured vs. predicted median relative distortion E, from the first-order least-squares model (dots) and from (7) (crosses); dashed: bisector. Right: measured and predicted E along the mission.

Remark 1 (Which tail to test): The collapse detectors of Section V threshold a statistic that decreases when the pairwise distances shrink—the signature of a single transmitter seen by all receivers [4, Result 1]. A ground array with one transmitter per satellite dilates the formation instead: the same statistic *increases*, and the attack sits in the tail these detectors never examine. The dilation is nevertheless a systematic bias of the distance ratios, and a two-sided test on an aggregate statistic is the natural tool to detect it.

V. Relative-Distance Detection Methods

All detectors operate on the normalized distance ratios

$$r_{ij}(t) = \frac{\hat{d}_{ij}(t)}{d_{ij}^{\mathrm{meas}}(t)}, \qquad \hat{d}_{ij} = \|\hat{\mathbf{p}}_i - \hat{\mathbf{p}}_j\|, \quad (8)$$

computed over all $\binom{m}{2} = 300$ pairs at every epoch, where d_{ij}^{meas} is the instantaneous UWB measurement of (2); under authentic conditions $r_{ij} \approx 1$.

A. Collapse tests (Jansen et al.)

The ACSAC 2016 detector [5] declares spoofing when the pairwise distances *shrink*:

$$f\big(\{r_{ij}\}\big) \underset{H_1}{\overset{H_0}{\gtrless}} \lambda, \qquad \text{alarm if } f < \lambda, \quad (9)$$

with the three variants of its Appendix A implemented here:

$$f_{\min} = \min_{i<j} r_{ij}, \quad f_{\max} = \max_{i<j} r_{ij}, \quad f_{\mathrm{maj}} = \operatorname*{med}_{i<j} r_{ij}, \quad (10)$$

where the median form is equivalent to the paper's normalized majority voting: at least half of the pairs below λ iff the median is. Jansen et al. set λ from a Rician (H_0) / Rayleigh (H_1) model of the pairwise distance with spatially correlated position noise, $\sigma_d = \sqrt{2}\,\sigma\sqrt{1-\rho}$; we calibrate λ empirically to the same effect (Section V-C).

B. Two-sided tests

Two-sided tests fire on any deviation of the distance ratios from one. The *worst-pair* test

$$T_{\mathrm{wp}} = \max_{i<j} \left| r_{ij} - 1 \right|, \qquad \text{alarm if } T_{\mathrm{wp}} > \lambda, \tag{11}$$

is the form used by the GNSS-vs-UWB check of [7] ($|d_{\mathrm{GPS}} - d_{\mathrm{UWB}}| > d_{\mathrm{thr}}$ on any pair) and, with temporal accumulation (CUSUM), by the WASP-aware countermeasure discussed in [9], which must cope with a residual whose sign varies from pair to pair. Because it is governed by the noisiest pair, it is sensitive to outliers and weak against a bias shared by all pairs. Against such a bias an *aggregate* statistic is more powerful—the χ^2 sum over double differences of [11], or simply the median—and the sign-coherent dilation of Section IV-C is exactly the case in which spatial aggregation pays: we therefore also evaluate the *two-sided majority* test

$$T_{\mathrm{maj}} = \left| \operatorname*{med}_{i<j} r_{ij} - 1 \right|, \qquad \text{alarm if } T_{\mathrm{maj}} > \lambda, \tag{12}$$

the two-sided counterpart of f_{maj} in (10), which is what Remark 1 calls for. Finally, since the dilation is persistent, both two-sided statistics are also evaluated after averaging over a sliding window of 12 epochs (1 min) within each mission, as an example of temporal integration.

C. Threshold calibration and metrics

Following the Neyman–Pearson practice of [5], thresholds are calibrated on the H_0 (no-spoofing) scenario simulated with the same noise levels: λ is the empirical P_{FA}-quantile (lower for the collapse tests, upper for the two-sided tests) of the statistic over all H_0 epochs, with $P_{\mathrm{FA}} = 10^{-2}$. Performance is reported as (i) the detection probability P_{D}, the fraction of attacked epochs that raise an alarm; (ii) ROC curves and their area (AUC) obtained by sweeping λ, oriented so that AUC $= 0.5$ is chance and AUC < 0.5 means that the attack moves the statistic to the side opposite to the alarm region; and (iii) the median relative distance distortion $E = \mathrm{med}_{i<j} |\hat{d}_{ij} - d_{ij}|/d_{ij}$, the geometric evidence available to any relative-distance detector. Because the epochs of one mission share clocks, trajectory and satellite geometry, the Monte Carlo unit is the *mission*: 95 % confidence intervals on P_{D} are obtained by a paired bootstrap over missions (500 resamples; H_0 and attack of a mission share the random seed, and the threshold is re-calibrated in every resample).

VI. Simulation Results

A. Setup and formation-spacing regime

Table I summarizes the parameters. We sweep the formation spacing over $g \in \{5, 8, 10, 12, 15, 20\}$ m. This range is the one of outdoor formation flight on GPS: experimental leader–follower formations of quadcopters are flown at 5 m spacing, with separation errors of 0.4 m [24] and of 1.4–1.9 m RMSE in a three-quadcopter experiment with RTK corrections;[1] a design method that derives the minimum safe spacing from the combined navigation and control error yields spacings of metres for GNSS-based navigation [25]; the multi-receiver prototype of [5] uses a 5 m square, and the swarm detector of [7] is designed around a 5 m GPS error budget. Closer spacings, down to about one metre in drone light shows,[2] require centimetre-level RTK positioning and are outside the standard-GPS regime considered here, while wider spacings belong to coverage and surveillance swarms rather than to formation flight. For each spacing we simulate $N_{\mathrm{MC}} = 20$ independent missions (random start time, clock biases and noise), each of 240 epochs, with and without the attack. The simulator and detection code are available in the companion repository.[3]

TABLE I
Simulation parameters.

Parameter	Value
Swarm size m	25 (5×5 grid)
Formation spacing g	5, 8, 10, 12, 15, 20 m
Mission duration / epoch	1200 s / $T_s = 5$ s
Satellites	24 (6 planes, 55°, 20 200 km)
Pseudorange noise σ_{pr}	3 m
Ranging noise σ_{rng}	0.1 m
Ground transmitters N_{sp}	24 on circle, $R_{\mathrm{sp}} = 10$ km
Transmitter EIRP	1 mW + 1 dB
Ground-to-UAV channel	Eq. (3), [10]
Spoofer power margin (all links)	13.2–29.0 dB
Largest transmitter elevation angle	6.0°
Target false-alarm rate P_{FA}	10^{-2}
Monte Carlo missions per spacing	20

B. Attack effectiveness

The attack succeeds: the centre UAV's estimated position is dragged by the full decoy offset, 4.0 km on average over the mission, at every epoch (Fig. 3, top), and every UAV in the swarm is captured on all its visible satellites. The bottom panel shows the price paid by a ground array: the relative-distance distortion is raised above the no-attack noise floor throughout the mission.

C. Formation distortion and dilation

Figure 4 reports the median relative distance distortion E over the 20 missions (band: 10th–90th percentile across missions). Under H_0 the distortion is pure GNSS noise and decays as $1/g$, from 22 % at $g = 5$ m to 4.6 % at 20 m. Under attack it decreases from 38 % at 5 m to 23 % at 20 m, i.e. towards the scale-free plateau of the residual of Section IV-C; the plateau is a median over missions whose geometric dilation ranges from 6.6 to 84 % (10th–90th percentiles at $g = 10$ m, noiseless), depending on the elevations of the satellites in view. Its sign is a dilation: the median distance ratio $\mathrm{med}\, r_{ij}$ is 1.23 under attack at $g = 20$ m against 1.01 without attack (1.37 against 1.16 at 5 m, where noise biases both upward),

[1] T. Grundler and J. Ryan, "Mesh-networked UAV swarm: experimental leader–follower formation control," senior project, Cal Poly, 2025.

[2] Show drones are flown at a minimum in-air spacing of 0.8 m with RTK positioning (vendor data, 2026).

[3] https://github.com/silviasch01/GPS_spoofing_distribuito

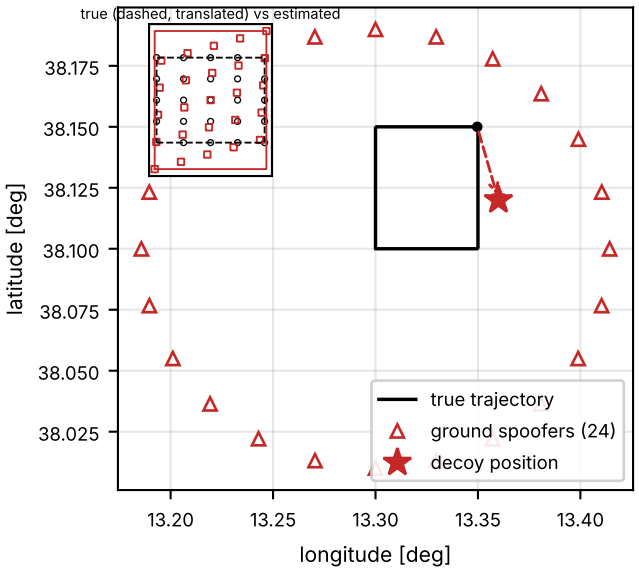


Fig. 2. Attack scenario (one epoch, $g = 10\,\mathrm{m}$). The swarm (true positions, dot) flies its mission inside a 10 km circle of 24 fixed ground transmitters (triangles), each replaying one satellite; the estimated centre (square) is displaced onto the decoy (star). Inset (noiseless run, same scale): the estimated grid (red squares) against the true grid translated by the same offset (black circles, dashed outline)—the formation is dilated (by 18 % at this epoch), not preserved.

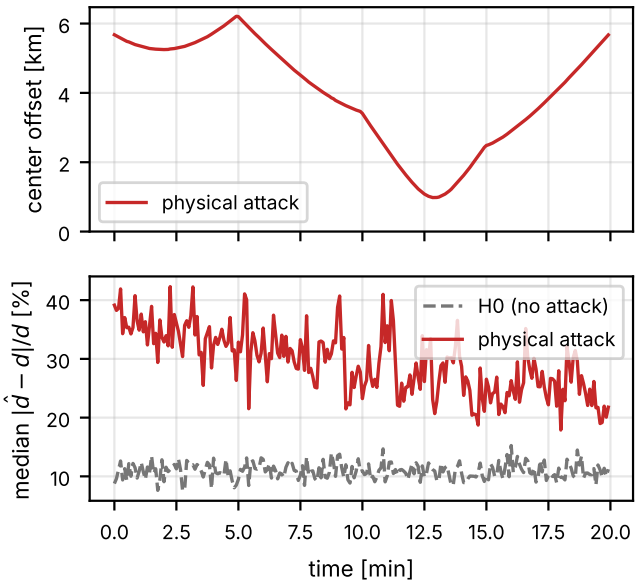


Fig. 3. Per-epoch behaviour at $g = 10\,\mathrm{m}$ (one mission, $\sigma_{\mathrm{pr}} = 3\,\mathrm{m}$). Top: position offset of the centre UAV. Bottom: median relative-distance error with and without the attack.

and it exceeds one in every attacked epoch. The computed formation is enlarged by about a quarter of the baseline; it never collapses.

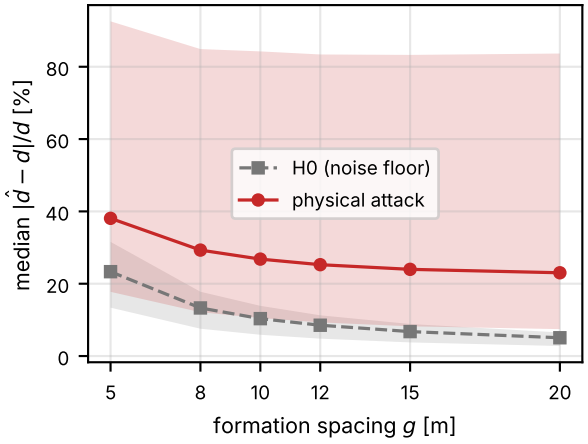


Fig. 4. Median relative-distance distortion vs. formation spacing (median over 20 missions; band: 10th–90th percentile across missions).

D. Detection performance

Table II and Fig. 5 report P_{D} at $P_{\mathrm{FA}} = 10^{-2}$ with 95 % confidence intervals, Table III the ROC areas, and Fig. 6 the ROC curves at $g = 10\,\mathrm{m}$. Three observations follow.

First, **the collapse detectors test the wrong tail**. All three Jansen variants give $P_{\mathrm{D}} \leq 0.011$ (95 % upper bound), at or below the false-alarm rate, at every spacing. Their AUC is *below* chance—down to 0.001 for the majority variant at 20 m—which does not mean that the attack is inseparable from authentic operation: the attacked statistic is *larger* than the authentic one in essentially every epoch, i.e. perfectly separable, but on the side these one-sided tests never examine (Remark 1). The majority variant, the one recommended in [5], never fires at all; no threshold choice changes this.

Second, **the worst-pair test is masked by noise**: P_{D} rises only from 0.03 at 5 m to 0.26 at 20 m (AUC 0.57–0.74), with wide confidence intervals that reflect the mission-to-mission variability of the dilation. The statistic (11) is governed by the noisiest of 300 pairs, whose H_0 tail at $\sigma_{\mathrm{pr}} = 3\,\mathrm{m}$ is wider than the dilation of the typical pair over the whole formation-flight regime; averaging it over a one-minute window raises P_{D} only to 0.12–0.38 (Table IV), still far from reliable detection.

Third, **the two-sided majority test sees the dilation**: it detects it reliably from 15 m ($P_{\mathrm{D}} = 0.85$ at 15 m, 0.98 at 20 m), in a majority of epochs at 8–12 m (0.57–0.71) and in more than a third of the epochs at 5 m (0.38), with AUC from 0.83 to 0.999. The aggregate statistic (12) averages the noise over all pairs and retains the common bias, which is exactly the signature left by a ground array. Its ROC at 10 m (Fig. 6) is the mirror image of the collapse majority test's: the same statistic, tested on the right side. With the same one-minute integration it becomes reliable over most of the regime: $P_{\mathrm{D}} = 0.52$ at 5 m, 0.80 at 10 m, 0.93 at 12 m and 1.00 from 15 m (Table IV).

TABLE II

DETECTION PROBABILITY AT $P_{\mathrm{FA}} = 10^{-2}$ AGAINST THE PHYSICAL DISTRIBUTED ATTACK ($\sigma_{\mathrm{pr}} = 3\,\mathrm{m}$, $\sigma_{\mathrm{rng}} = 0.1\,\mathrm{m}$; 20 MISSIONS, 4800 EPOCHS PER CELL; 95 % CI FROM THE PAIRED MISSION BOOTSTRAP).

g [m]	Jansen min	Jansen max	Jansen majority	Worst-pair	Two-sided majority
5	0.005 [0.004, 0.011]	0.007 [0.004, 0.009]	0.000 [0.000, 0.000]	0.029 [0.016, 0.048]	0.382 [0.215, 0.569]
8	0.002 [0.000, 0.004]	0.003 [0.002, 0.005]	0.000 [0.000, 0.000]	0.065 [0.024, 0.119]	0.572 [0.386, 0.780]
10	0.002 [0.000, 0.004]	0.002 [0.001, 0.005]	0.000 [0.000, 0.000]	0.113 [0.044, 0.195]	0.632 [0.461, 0.823]
12	0.001 [0.000, 0.005]	0.001 [0.000, 0.003]	0.000 [0.000, 0.000]	0.153 [0.064, 0.268]	0.713 [0.577, 0.871]
15	0.002 [0.000, 0.004]	0.001 [0.000, 0.001]	0.000 [0.000, 0.000]	0.193 [0.088, 0.335]	0.845 [0.729, 0.947]
20	0.001 [0.000, 0.003]	0.000 [0.000, 0.001]	0.000 [0.000, 0.000]	0.262 [0.131, 0.426]	0.975 [0.946, 0.997]

TABLE III

ROC AREA (AUC); 0.5 IS CHANCE, VALUES BELOW 0.5 MEAN THE ATTACK MOVES THE STATISTIC TO THE SIDE OPPOSITE TO THE ALARM REGION (SEPARABLE, BUT FROM THE OTHER TAIL).

g [m]	J. min	J. max	J. maj.	Worst-pair	2-sided maj.
5	0.312	0.430	0.171	0.570	0.829
8	0.262	0.378	0.072	0.622	0.928
10	0.221	0.349	0.037	0.651	0.963
12	0.189	0.324	0.019	0.676	0.981
15	0.156	0.293	0.006	0.706	0.994
20	0.122	0.255	0.001	0.743	0.999

TABLE IV

TWO-SIDED TESTS WITH 1-MIN TEMPORAL INTEGRATION (MOVING MEAN OVER 12 EPOCHS): P_{D} AT $P_{\mathrm{FA}} = 10^{-2}$, 95 % CI.

g [m]	Worst-pair, 1-min	Two-sided majority, 1-min
5	0.117 [0.019, 0.205]	0.521 [0.292, 0.701]
8	0.158 [0.055, 0.309]	0.698 [0.491, 0.851]
10	0.182 [0.059, 0.308]	0.804 [0.649, 0.945]
12	0.224 [0.079, 0.382]	0.932 [0.832, 0.991]
15	0.280 [0.102, 0.454]	1.000 [0.994, 1.000]
20	0.376 [0.167, 0.602]	1.000 [1.000, 1.000]

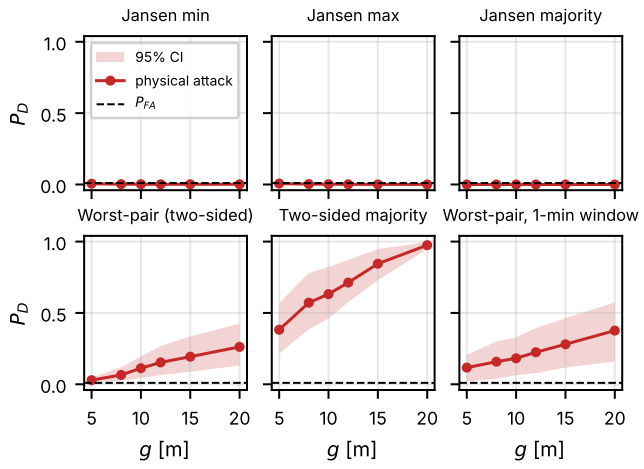


Fig. 5. Detection probability vs. formation spacing at $P_{\mathrm{FA}} = 10^{-2}$ (20 missions; band: 95 % bootstrap CI). The collapse detectors never leave the P_{FA} line; the two-sided majority test detects the dilation at every spacing.

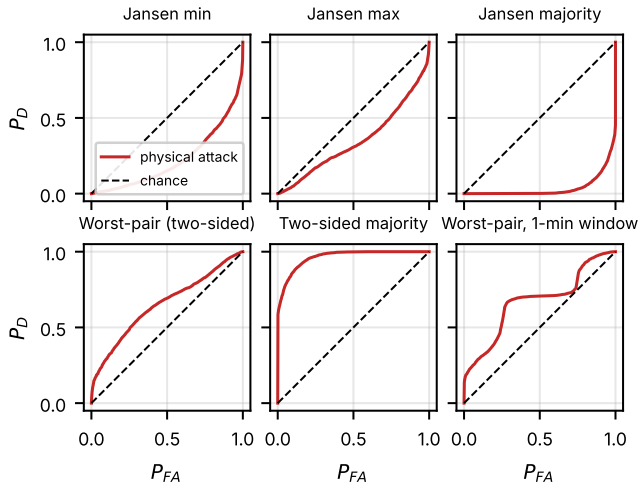


Fig. 6. ROC curves at $g = 10\,\mathrm{m}$ against the physical distributed attack. The collapse detectors operate below chance (separable from the opposite tail); the two-sided majority test operates near the top-left corner.

E. The cost in altitude: from a ground array to a WASP-like constellation

Equation (6) predicts that the dilation decays as the transmitters are raised towards the satellite elevations and vanishes only for transmitters on the satellites' lines of sight—the placement realized at altitude by GNSS-WASP [9]. We quantify this trade-off with two experiments at $g = 10\,\mathrm{m}$, the centre of the formation-flight regime, on the same 20 missions (same seeds) as the main results. The dilation is a property of the geometry, so it is measured on noiseless runs of every mission; the noisy runs serve only to measure P_{D}. Elevated transmitters use a free-space link (Section III) with the EIRP raised to 10 mW: capture is complete in every mission (minimum margin 13.9 dB), and since the delays do not depend on the channel the ground point of the sweep coincides with the main results.

Altitude sweep. The 24-transmitter, 10 km array is raised from the ground to 10 km, i.e. from $\theta_k \approx 0°$ to $47°$ as seen from the swarm (Fig. 7). The model reproduces the measured distortion of every mission at every altitude (top panel). The *signed* dilation, median over the 20 missions, decreases monotonically from $+22\,\%$ on the ground to $+15.5\,\%$ at $19°$, $+8.8\,\%$ at $27°$, crosses zero at $\theta_k \approx 36°$ (7 km of altitude

for this array) and turns into a *contraction* of $-15\,\%$ at $47°$: with a single altitude for all transmitters the residual cannot vanish for every satellite at once, so the satellites below θ_k pull the formation inward while those above still push it outward. The mission-to-mission spread is large—at the crossing the 10th–90th percentiles of the signed dilation are $-15\,\%$ and $+62\,\%$, because the balance depends on the elevations of the satellites in view—so the distortion E pooled over missions has its minimum ($10.7\,\%$) before the crossing, at $27°$, and is $14.6\,\%$ at the crossing itself.

The detectors follow the sign of the residual (middle panel): the two-sided majority test decreases from $P_{\mathrm{D}} = 0.63$ on the ground to a minimum of 0.39 at the crossing and rises again to 0.59 at $47°$ (with one minute of integration $0.80 \to 0.53 \to 0.71$); the collapse test of Jansen et al., blind on the ground, starts firing as the formation contracts ($P_{\mathrm{D}} = 0.22$ at $27°$, 0.44 at the crossing, 0.64 at $47°$); the worst-pair tests stay at 0.05–0.18 throughout, since the residual that survives at the crossing (10–15 % of the baseline, of varying sign across pairs and missions) remains below their noise. The minimum of the aggregate test is not lower because the crossing altitude is mission-dependent: at any single altitude most missions are still dilated or already contracted. The bottom panel makes this explicit by plotting, for every mission and configuration, P_{D} against the mission's own geometric dilation D. The two-sided majority test is blind in the window $D \in [-8\,\%, +5\,\%]$ ($P_{\mathrm{D}} < 0.1$) and detects outside it (0.71 for $D = +12$–$20\,\%$, 0.93 beyond $+20\,\%$, 0.81 beyond $-20\,\%$); the collapse test is its mirror image on the contraction side. The window is shifted towards contraction rather than centred on zero, and the shift is the noise bias: under H_0 the median ratio is $1+b_0$ with $b_0 \approx 0.045$ at 10 m (noise lengthens distances on average), under attack it is approximately $(1+D)(1+b_0/(1+D)^2) \approx 1+D+b_0$, so the attacked statistic $|D+b_0|$ falls within the H_0 spread for D slightly negative—a contraction of a few percent reads as no attack. The attacker's optimum with a common-altitude array is therefore a narrow, mission-dependent band around the zero crossing (about 7 km of altitude for a 10 km array), not the highest altitude; reaching it for every mission would require per-satellite altitudes, i.e. the line-of-sight placement.

Line-of-sight placement. The second experiment is a *static* version of the GNSS-WASP geometry: one transmitter per satellite visible at the start of the mission, placed on its line of sight from the array centre at slant range R_{los} (altitude $R_{\mathrm{los}} \sin \mathrm{el}_s$, up to 9 km for $R_{\mathrm{los}} = 10$ km). Since such a constellation is on the lines of sight only for a swarm near the reference point—as in GNSS-WASP, whose bias grows with the distance from it—the experiment uses a loitering mission confined within 700 m of the centre (`traj_scale`$= 0.1$), and compares with the ground circle on the same mission (Table V). The line-of-sight placement cancels the signed dilation ($+20.5\,\%$ for the ground circle, $+5.0$, $+0.9$ and $+0.3\,\%$ at $R_{\mathrm{los}} = 2$, 5 and 10 km) but leaves a residual distortion of 11, 5.8 and $4.3\,\%$ of varying sign, because the transmitters are fixed while the satellites move and new ones rise during the 20-minute mission. The two-sided majority test drops from $P_{\mathrm{D}} = 0.62$ (AUC 0.96) to 0.29, 0.23 and 0.18 (AUC 0.75, 0.65, 0.60), the worst-pair test to 0.04, and the collapse test fires in about a tenth of the epochs on the contracting missions. GNSS-WASP's airborne transmitters track the satellites, so this static residual is an upper bound of the WASP error and its detectability an upper bound of WASP's: a constellation on the lines of sight at slant ranges of 5 km and beyond is detected in fewer than a quarter of the epochs by the best relative-distance test at this spacing, against two thirds for the ground array—the price being airborne platforms at several kilometres of altitude.

TABLE V
LOITERING MISSION ($g = 10$ M, 20 MISSIONS): GROUND CIRCLE VS. STATIC LINE-OF-SIGHT (WASP-LIKE) PLACEMENT AT SLANT RANGE R_{los}. GEOMETRIC DILATION D AND DISTORTION E (NOISELESS); P_{D} AT $P_{\mathrm{FA}} = 10^{-2}$ WITH NOISE (AUC IN PARENTHESES).

Placement	D	E	Jansen maj.	Worst-pair	2-sided maj.
Ground circle, 10 km	$+20.5$	20.5	0.00 (0.04)	0.11 (0.65)	0.62 (0.96)
LOS, $R_{\mathrm{los}} = 2$ km	$+5.0$	11.0	0.13 (0.34)	0.04 (0.56)	0.29 (0.75)
LOS, $R_{\mathrm{los}} = 5$ km	$+0.9$	5.8	0.10 (0.41)	0.04 (0.54)	0.23 (0.65)
LOS, $R_{\mathrm{los}} = 10$ km	$+0.3$	4.3	0.09 (0.44)	0.04 (0.53)	0.18 (0.60)

VII. DISCUSSION AND LIMITATIONS

What a ground array can and cannot do: A distributed ground spoofer defeats the premise of the classical multi-receiver countermeasure—it does not collapse the formation—but it cannot preserve it either: the elevation mismatch of (6) dilates the computed formation by about $1/\langle \cos \mathrm{el} \rangle - 1 \approx 23\,\%$ of the baseline, independently of transmitter power and of the formation spacing. The practical consequence for the defender is a change of signature, not a loss of observability: collapse tests should be replaced, or at least complemented, by two-sided aggregate tests on the distance ratios, which detect the dilation reliably at 15–20 m and in a majority of epochs at 5–10 m with standard positioning, and reliably from 12–15 m with one minute of integration. Temporal integration helps the worst-pair test but does not make it competitive with the aggregate statistic. Lower-noise fixes (RTK) or wider formations raise all two-sided detectors, since the dilation is scale-free (about 23 % of the baseline) while the noise floor decays as $1/g$: in supplementary runs at 50 and 100 m spacing (20 missions each, same attack model and seeds, reproduced bit for bit by the released code) the two-sided majority test reaches $P_{\mathrm{D}} = 1.00$, the worst-pair test 0.56 and 0.65 (AUC 0.86 and 0.96), and the collapse detectors remain at zero. Raising the transmitters reduces and then reverses the dilation (Section VI-E); a defender should therefore test the distance ratios on *both* sides, which the two-sided majority statistic does by construction, and should expect a blind window of about $[-8, +5]\,\%$ of geometric dilation, shifted towards contraction by the noise bias of the median ratio. The design of a spoofer that compensates the residual with per-satellite altitudes is left to future work.

Scope of the result: The evaluation covers detectors that are functions of relative position/distance quantities, over the

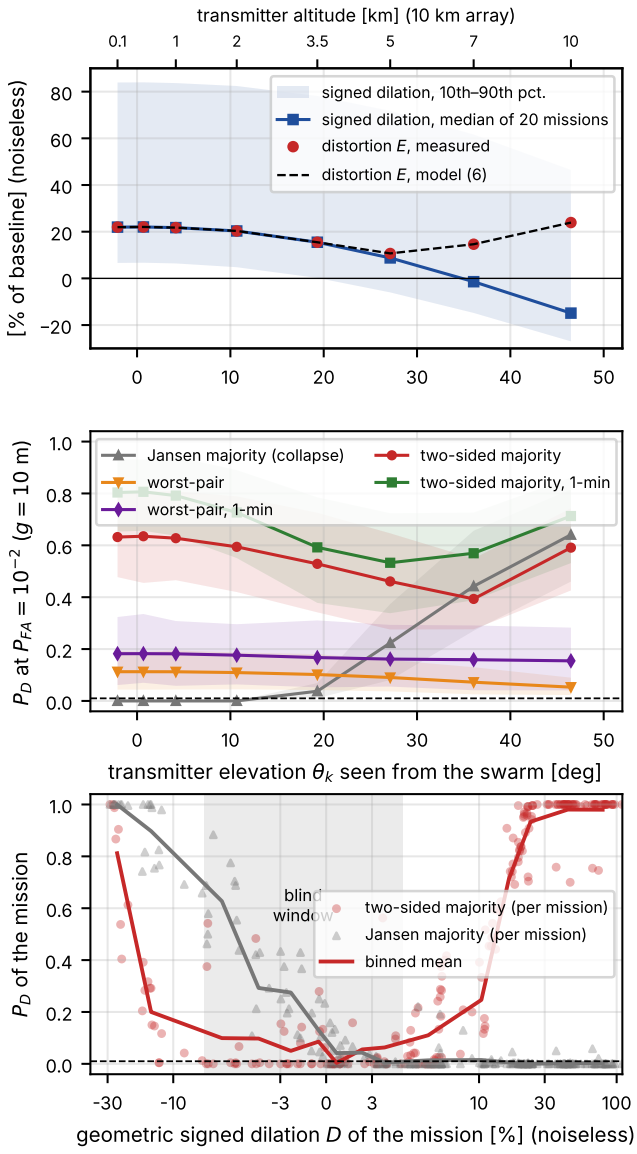


Fig. 7. Transmitter altitude sweep (24-transmitter, 10 km array, $g = 10$ m, 20 missions). Top: signed dilation (median and 10th–90th percentiles over the missions) and distortion E (measured, and from the model (6)), all from noiseless runs. Middle: P_{D} at $P_{\mathrm{FA}} = 10^{-2}$ with noise (bands: 95 % CI). Bottom: P_{D} of each mission against its geometric dilation, for all altitudes and placements; the two-sided majority test is blind for $D \in [-8, +5]$ %, the collapse test only fires for contractions.

spacing of GPS-based formation flight. It does *not* extend to (i) *direction-of-arrival* detectors: a circle of ground transmitters produces arrivals at near-zero elevation, incongruent with the satellite constellation, so multi-antenna arrays with attitude information [2], [15] would see our attacker, unless its transmitters are placed along the satellite directions as in [9]; nor to (ii) *absolute-reference* checks—ranging to fixed infrastructure, TDoA from ground sensor networks [13], terrain or signal-of-opportunity positioning—for which the 4 km displacement is directly observable; nor to (iii) GNSS/INS detectors such as SemperFi [16] and INS innovation monitors [17], [18], which respond to the time variation of the induced error and were not evaluated.

Relation to formation-preserving attacks: Tippenhauer et al. [4] prove that a formation-preserving placement exists and GNSS-WASP [9] realizes it with transmitters in line with the satellites at altitude; Park and Yoo [21] show that an exactly common translation is invisible to any relative-only detector. Our attacker is deliberately more constrained, and the quantity we measure and model is what separates it from those ideal attacks: a dilation large enough to be detected by the right test, and the reason why detectors tuned to the single-transmitter signature miss it. Section VI-E connects the two regimes explicitly: raising a common-altitude array does not remove the residual but changes its sign, so the attacker's best altitude is a mission-dependent band around the zero crossing (about 7 km for a 10 km array), and only a constellation on the lines of sight at several kilometres of slant range approaches the undetectable regime of [9]; even then a static constellation leaves a 4–6 % residual over a 20-minute mission because the satellites move while the transmitters do not.

Deployment, steady state, and other simplifications: The result assumes an array that encloses the swarm's operating area (Section IV-C); a static array that the swarm leaves deforms the formation by more than 50 % and is readily detected. The decoy position is constant, so the attack is modeled in its steady state after capture; the transient at capture and time-varying decoys are not modeled. The simulator uses a spherical Earth, a synthetic constellation, uncorrelated pseudorange noise (spatial correlation would lower the detector's noise floor [5] and favour the two-sided tests), no multipath or shadowing, and signal capture by power comparison without tracking-loop dynamics. Twenty missions per spacing bound P_{D} to within about ± 0.1 for the two-sided tests; the conclusions on the collapse detectors do not depend on this resolution.

VIII. Conclusion

We simulated a physical distributed ground spoofer—24 fixed transmitters, one impersonated satellite each, real propagation and channel—attacking a UAV swarm in the 5–20 m spacing of outdoor GPS-based formation flight, and evaluated the classical relative-distance detectors against it. The attack displaces the swarm by 4 km but cannot preserve its formation: the mismatch between satellite and transmitter elevations dilates the computed formation by about 23 % of the baseline, a value reproduced epoch by epoch by a first-order model. The dilation lies in the tail that the collapse detectors of Jansen et al. never test ($P_{\mathrm{D}} \leq 0.011$, AUC down to 0.001) and is largely masked by GNSS noise for the worst-pair GNSS-vs-UWB check ($P_{\mathrm{D}} \leq 0.26$), but a two-sided test on the median distance ratio detects it reliably from 15 m and in a majority of epochs from 8 m (P_{D} from 0.38 at 5 m to 0.98 at 20 m). Raising the transmitters does not remove the residual but reverses it: the attacker's best altitude is a narrow, mission-dependent band around the zero crossing of the dilation (about 7 km for a 10 km array), where the aggregate

test is blind, and only a constellation on the satellites' lines of sight at several kilometres of slant range comes close to undetectability. Against distributed ground spoofers, relative-distance detection should therefore test for dilation rather than collapse—on both sides—and be complemented by direction-of-arrival or absolute-reference information for the airborne attackers that can forge the elevations too.

Acknowledgments

[TODO: ringraziamenti, funding]

References

[1] T. E. Humphreys, B. M. Ledvina, M. L. Psiaki, B. W. O'Hanlon, and P. M. Kintner, Jr., "Assessing the spoofing threat: Development of a portable GPS civilian spoofer," in *Proceedings of the 21st International Technical Meeting of the Satellite Division of The Institute of Navigation (ION GNSS 2008)*, Savannah, GA, USA, 2008, pp. 2314–2325.

[2] M. L. Psiaki and T. E. Humphreys, "GNSS spoofing and detection," *Proceedings of the IEEE*, vol. 104, no. 6, pp. 1258–1270, 2016.

[3] A. Altaweel, H. Mukkath, and I. Kamel, "GPS spoofing attacks in FANETs: A systematic literature review," *IEEE Access*, vol. 11, pp. 55 233–55 280, 2023.

[4] N. O. Tippenhauer, C. Pöpper, K. B. Rasmussen, and S. Čapkun, "On the requirements for successful GPS spoofing attacks," in *Proceedings of the 18th ACM Conference on Computer and Communications Security (CCS)*. ACM, 2011, pp. 75–86.

[5] K. Jansen, N. O. Tippenhauer, and C. Pöpper, "Multi-receiver GPS spoofing detection: Error models and realization," in *Proceedings of the 32nd Annual Conference on Computer Security Applications (ACSAC)*. ACM, 2016, pp. 237–250.

[6] P. F. Swaszek and R. J. Hartnett, "Spoof detection using multiple COTS receivers in safety critical applications," in *Proceedings of the 26th International Technical Meeting of the Satellite Division of The Institute of Navigation (ION GNSS+ 2013)*, Nashville, TN, USA, 2013, pp. 2921–2930.

[7] P. Mykytyn, M. Brzozowski, Z. Dyka, and P. Langendoerfer, "GPS-spoofing attack detection mechanism for UAV swarms," in *2023 12th Mediterranean Conference on Embedded Computing (MECO)*, 2023.

[8] H. Bi, X. Li, J. Hu, W. Ni, Y. Wang, and X. Wang, "Detection and mitigation of position spoofing attacks on cooperative UAV swarm formations," *IEEE Transactions on Information Forensics and Security*, vol. 19, pp. 1883–1895, 2024, arXiv:2312.03787; DOI to be added.

[9] C. Tibaldo, H. Sathaye, G. Camurati, and S. Čapkun, "GNSS-WASP: GNSS wide area SPoofing," in *Proceedings of the 34th USENIX Security Symposium*. USENIX Association, 2025, pp. 7039–7058.

[10] A. Al-Hourani and K. Gomez, "Modeling cellular-to-UAV path-loss for suburban environments," *IEEE Wireless Communications Letters*, vol. 7, no. 1, pp. 82–85, 2018.

[11] N. Stenberg, E. Axell, J. Rantakokko, and G. Hendeby, "Results on GNSS spoofing mitigation using multiple receivers," *NAVIGATION: Journal of the Institute of Navigation*, vol. 69, no. 1, p. navi.510, 2022.

[12] P. F. Swaszek, R. J. Hartnett, and K. C. Seals, "GNSS spoof detection using independent range information," in *Proceedings of the 30th International Technical Meeting of the Satellite Division of The Institute of Navigation (ION GNSS+ 2017)*, Portland, OR, USA, 2017, pp. 2838–2853, title field unverified.

[13] K. Jansen, M. Schäfer, D. Moser, V. Lenders, C. Pöpper, and J. Schmitt, "Crowd-GPS-Sec: Leveraging crowdsourcing to detect and localize GPS spoofing attacks," in *2018 IEEE Symposium on Security and Privacy (SP)*, 2018, pp. 1018–1031.

[14] A. Kalantari and E. G. Larsson, "Statistical test for GNSS spoofing attack detection by using multiple receivers on a rigid body," *EURASIP Journal on Advances in Signal Processing*, vol. 2020, no. 5, 2020, article number unverified (5 or 8).

[15] D. Borio and C. Gioia, "A sum-of-squares approach to GNSS spoofing detection," *IEEE Transactions on Aerospace and Electronic Systems*, vol. 52, no. 4, pp. 1756–1768, 2016.

[16] H. Sathaye, G. LaMountain, P. Closas, and A. Ranganathan, "SemperFi: Anti-spoofing GPS receiver for UAVs," in *Proceedings of the Network and Distributed System Security Symposium (NDSS)*, 2022, dOI unverified; preprint arXiv:2105.01860.

[17] C. Tanil, S. Khanafseh, M. Joerger, and B. Pervan, "An INS monitor to detect GNSS spoofers capable of tracking vehicle position," *IEEE Transactions on Aerospace and Electronic Systems*, vol. 54, no. 1, pp. 131–143, 2018.

[18] B. Kujur, S. Khanafseh, and B. Pervan, "Detecting GNSS spoofing using temporal behavior of innovation sequences," *NAVIGATION: Journal of the Institute of Navigation*, vol. 71, no. 1, p. navi.629, 2024, title field unverified.

[19] M. Ceccato, F. Formaggio, and S. Tomasin, "Spatial GNSS spoofing against drone swarms with multiple antennas and Wiener filter," *IEEE Transactions on Signal Processing*, vol. 68, pp. 5782–5794, 2020.

[20] Y. Chen, X. Dong, and R. Duan, "GPS swarm spoofing: Theoretical analysis and practical solutions," in *Proceedings of the IEEE International Conference on Communications (ICC)*, 2025, pp. 3345–3350, title field unverified.

[21] M. Park and J. S. Yoo, "Rigid-covert GNSS spoofing of UAV swarms: A structural blind spot, its detection limit, and absolute-anchor defenses," arXiv:2608.06885, 2026, preprint, not peer-reviewed.

[22] J. Aspnes, T. Eren, D. K. Goldenberg, A. S. Morse, W. Whiteley, Y. R. Yang, B. D. O. Anderson, and P. N. Belhumeur, "A theory of network localization," *IEEE Transactions on Mobile Computing*, vol. 5, no. 12, pp. 1663–1678, 2006.

[23] A. Sebastian and B. C. Manjith, "A comprehensive survey of GPS spoofing attack detection techniques in UAV systems," in *2026 IEEE International Conference on Intelligent Technologies, Systems and Innovation Frontiers (ICITSIF)*, 2026, pp. 1041–1046, author first names and conference expansion unverified.

[24] J. H. Cheok, V. R. Aparow, J. Z. N. Ng, J. L. Cheah, and D. Leong, "Development and optimization of formation flying for unmanned aerial vehicles using particle swarm optimization based on reciprocal velocity obstacles," *SAE International Journal of Aerospace*, vol. 15, no. 2, pp. 171–184, 2022, author list to be confirmed on the DOI page.

[25] P. Guo, R. Zhang, and B. Xu, "Safety separation distance design for UAV formation based on system performance," *IEEE Transactions on Aerospace and Electronic Systems*, vol. 61, no. 3, pp. 5980–5995, 2025, dOI to be added.